15-1-25

- Baudhayana (~800 BC)
  - approx of $\sqrt{2}$
  - version of pythagoras Theorem
- Katyayana (~300 BC)
  - Vartika, Vyakarana, Sulba Sutras
- Pingala (~500 BC)
  - Matrameru, Binary numeral system, Arithmetical Δ^{le}
- Aryabhata (476 - 550 CE)
  - Expln of lunar & solar eclipse
  - Rotn of Earth on axis
  - Reflctn of light by moon
  - Sinusoidal fns.

- Varahamihara (505 - 587 CE)
  - Pancha - siddhānti kā
  - Brihat - Samhita
  - Brihat Jataka
- Bhaskara II (1114 - 1185 CE)
  - principles of diff^n calculus
  - & its appl^n to astronomical problems & computations.
- Srinivasa Ramanujan (1887 - 1920).
  - Mock θ f^n
  - Ramanujan conjecture
  - 's sum
- Prasanta C Mahalanobis (1893 - 1972)
  - Mahalanobis distance
  - " model
  - Founder of ISI in Kolkatta

- Charles Babbage, John Von Neumann, Konrad Zuse, Alan Turing. → difference machine (1812)

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* 300 BC → Abacus
1947 → Transistor
1968 → Intel
1977 → Apple
1981 → IBM PC
1991 → World Wide Web

History of computer.

(Prehistory → Electronics → Mini → Micro → Network)
→ counting fingers, pebbles, # on walls, # on bone, # on sand.

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* Premechanical computing — Before 3000 BCE
Mechanical computing — After 3000 BCE

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* Mechanical computers:
→ Abacus (3000 BC): Slide beads up & down on the rods to '+' or '−'
→ Napier's Bones & logarithms
→ slide Rule Oughtred's (1621) & Schickard's (1623)

↳ works on basis of log
↳ mechanical analog computer.

→ Pascal's Pascaline (1645)
↳ resembled a mech. calculator of 1940's
↳ can add & '−' by simple rotⁿ of dials on the machine's face.

→ Gottfried Wilhelm von Leibnitz's Stepped Reckoner (1694)
↳ could +, −, /, ×
↳ mechanical calculator.

→ Joseph Marie Jacquard's Punched card Controlled Looms (1804)
↳ used punched cards
↳ model for i/p & o/p

→ Charles Babbage
↳ Diff engine → to compile mathematical tables
↳ Analytical engine → incorporated ALU, CU in the form of conditional branching & loops & integrated memory.

→ Herman Hollerith & his Census Tabulating Machine (1884)
↳ electromechanical machine designed to assist in summarizing information stored on punched cards.
↳ was used to help process data for 1890 US census
↳ for accounting & analytical purposes.

→ Harvard Mark I (1944) aka IBM's Automatic Sequence (ASCC)

Imp Machines
IBM 700 Series
" 360/370 "

CDC star
Cray 1
DECVAX
Apple Macintosh

ENIAC → Electronic Numerical Integrator Analyser & computer.
↳ developed by army ordinance to compute WW-II ballistic firing tables.
↳ weighed 30 tons & consumed 200 KW of power.
↳ developed by Ballistic Research facility.

1947 – Bell Labs is the birthplace of transistor.
↓
regulates voltage / current to act as switch or gate.

UNIVAC → Universal Automatic computer.
UNIVAC → 16000 lbs, 5000 vaccumm tubes, &
↳ could perform 1000 calculat's / sec.
↳ could store 12000 digits in RAM.

EDVAC → Elect Discrete Variable computer
↳ simple binary comp with +, −, /, *

IC → Texas Instruments.
↳ made of semiconductor material.
All comp s can be made using chips instead of transistors.

→ IBM 360 Standard Institutional Mainframe comps.
↳ first family.

Ivan Sutherland - 1962 - Sketchpad.
Doug Engelbarts → invented computer mouse.

Moore's law → no. of components in an IC will x2 every year.

Intel 4004 → 6000 operat^ns/sec. It consists of 2,300 transistors & has a 10 mm^2 area.

8008 → first 8-bit microprocessor.

Pong → first computer games → tennis like 2 paddles & a ball.

PL/M → Programming lang / Microprocessor. to run on IBM 360

CP/M (Control program / Monitor) a simple OS in his PL/M language.

1972 - Gary Kildall writes the first SW for a microprocessor.

Visicalc - 1979 - Spreadsheet

Timesharing - the concept of linking a large number of users to a single computer via remote terminals is developed at MIT.

Packet Switching Network.
ARPANET.
World Wide Web.

1973 → basic ideas of Internet
1982 → TCP/IP as standard for ARPANET
1991 → WWW
CERN → releases the first Web Server.

5 Generations of Comp's

Gen	Core component	Time period
1st	Vacuum Tubes	1940 - 1956
2nd	Transistors	56 - 63
3rd	ICs	64 - 71
4th	VLSI	71 - 85
5th	ULSI	85 - present

divided into gens depending on
- hardW & softW
- speed, no of instructions /sec
- physical appearance
- Applications

resulting in better, cheaper & smaller comps that are more powerful & efficient.

1st Gen: UNIVAC, ENIAC
- vacuum tubes for circuitry
- magnetic drum for memory
- 400 bits
- HW progs in comp

} features.

Adv: only electronic device, first device to hold memory.

2nd Gen: Transistors
- Cryptic bin. machine to symbolic
- HLLs like COBOL & FORTRAN
- 1st devices to store instructions in memory.

Features: Transistors, Magnetic cores, 32k bits, Punch cards
Manufacturers: Control Data Corporation, (CDC), GE, IBM

Adv : size ↓ speed ↑ suitable
disadv : overheated, maintenance problems.

3rd Gen : → IC
↳ Transistors were miniaturized &

Manufacturers
IBM, NCR
honeywell

Features : silicon chips (ICs)
costs IC
128000 bits
Keyboards Entry.

Adv : size ↓, speed ↑ cheap
disadv : ICs are sophisticated.

4th Gen : → microprocessor
↳ Intel 4004 chip had all components of comp.
Features : LSI, ICs, 100 million bits, Reads programs off disks.

5th Gen : → Based on AI
↳ Voice Recognition.
↳ parallel processing
↳ natural language i/p
↳ capable of learning & self-organization.

Features : VLSI
Unlimited
CD-Rom, optical disk.

Next & 6th Gen :
Quantum computing Era :
HPC, Advanced AI, Neuromorphic computing.

Taxonomy of Comps : Mainframes, Minis, Micros, Supercomputers
Embedded.